

accuracy. All sensors require linearisation but to a different degree.

Signal conditioning is required with sensors as most of these provide low-level and non-linear outputs. To effectively and accurately measure signals, most analogue signals require some form of conditioning before these can be digitised and sent for further processing to the controller. For instance, filtering and amplification can dramatically improve the accuracy of thermocouple measurements. It is important that the required accuracy and resolution for an application is matched to the data acquisition and signal conditioning hardware that you select.

In the past, many complex analogue signal conditioning and calibration circuits were used that required manual calibration and precision resistors. Today, however, a digital design is used instead. Sensor outputs are often digitised by a high-resolution analogue-to-digital converter (ADC), which allows linearisation and calibration in software. This allows minimum operator involvement and also reduces cost and complexity.

Sensor excitation is another important concern. For instance, as RTDs and thermistors are resistive devices, you must supply these with an excitation current and then read the voltage across their terminals. If extra heat due to this cannot be dissipated, self-heating of the sensor is caused by the excitation current, which raises the temperature of the sensing element above the ambient temperature, thereby introducing errors in measurement. Power regulation is thus important to limit self-heating effects.

Stability defines how consistently a sensor maintains its accuracy over time. Generally, stability gets worsened by exposure to high temperatures. Wire-wound platinum and glass-encapsulated thermistors are the most stable, whereas thermocouples and silicon sensors are the least.

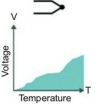
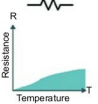
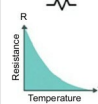
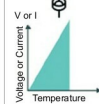
Thermocouple	RTD	Thermistor	IC Sensor
			
Advantages • Self-powered • Simple • Rugged • Inexpensive • Wide variety of physical forms • Wide temperature range	• Most stable • Most accurate • More linear than thermocouple	• High output • Fast • Two-wire ohms measurement	• Most linear • Highest output • Inexpensive
Disadvantages • Non linear • Low voltage • Reference required • Least stable • Least expensive	• Expensive • Slow • Current source required • Small resistance change • Four-wire measurement	• Non-linear • Limited temperature range • Fragile • Current source required • Self-heating	• $T < 250^{\circ}\text{C}$ • Power supply required • Slow • Self-heating • Limited configurations

Fig. 2: Advantages and disadvantages of the different temperature-monitoring sensors

Temperature range varies for each sensor type with the thermocouple family possessing the widest temperature range, spread across multiple thermocouple types.

Electrical noise can induce errors in temperature sensing. Mostly thermocouples and sometimes thermistors with very high resistances present this problem.

Sensor outputs vary by type of sensor. Most thermistors change resistance inversely-proportionally with temperature, thus the name negative temperature coefficient (NTC). Base metals such as platinum have positive temperature coefficient (PTC). Thermocouples have low milli-volt outputs that change with temperature. Silicon sensors are typically conditioned and come with a variety of digital outputs.

Cost is a major deciding factor. Although thermocouples are the least expensive and the most widely-used sensors, an NTC thermistor

generally provides the greatest value for its price.

Sensitivity allows a sensor to sense very small changes in temperature.

Resolution dictates the smallest detectable change in temperature that a system can detect.

Flexibility allows a sensor to be configured into a wide variety of physical forms such as very small packages.

Response time determines how quickly a system can react to any change in temperature. Generally, the smaller the sensor, the faster the response time.

Thermocouples

Thermocouples are by far the most widely-used type of temperature sensors in the industry. Made by joining two wires of dissimilar metals, a thermocouple takes advantage of the voltage induced at the point of contact between the wires when these